

Q10 Ans: B

Explanation

amplitude $x_0 = 4.0 \text{ m}$

max speed $v_0 = \omega x_0$

$$20 = \omega \times 4.0$$

$$\text{so } \omega = 5.0$$

$$\text{period } T = \frac{2\pi}{\omega} = \frac{2\pi}{5.0} = 1.257 \text{ s}$$

$$\text{time for speed to reduce from max to zero} = \frac{1}{4}T = \frac{1}{4} \times 1.257 = 0.31 \text{ s}$$



